

AI Learning Roadmap (No Data Science)

Phase 1: AI Fundamentals

- **What is AI?**

Understand AI as systems that mimic human intelligence for tasks like reasoning, learning, and perception.

- **Subsets of AI**

Explore key branches: Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), and Large Language Models (LLMs).

Phase 2: Machine Learning Basics

- **Types of ML**

Supervised (labeled data), Unsupervised (unlabeled data), and Reinforcement Learning (RL, agent-environment interaction).

- **Start with Supervised Learning**

- Classification (predicting categories) vs. Regression (predicting continuous values).
- Linear Regression (simple straight-line predictions).
- Logistic Regression (binary classification).
- KNN (K-Nearest Neighbors).
- Decision Trees, Random Forest (ensemble of trees), Bagging, Boosting.
- SVM (Support Vector Machines) and the Kernel Trick (handling non-linear data).

- **Feature Engineering Basics**

Scaling (normalization/standardization), encoding categorical data.

- **Overfitting/Underfitting**

Detection and fixes like regularization (L1/Lasso, L2/Ridge). #optional can skip

Phase 3: Unsupervised Learning

- **Key Concepts**

Types of clustering, PCA (Principal Component Analysis), SVD (Singular Value Decomposition), Curse of Dimensionality, Anomaly Detection Algorithms (awareness of isolation forests, etc.).

- **Clustering Algorithms**

- K-Means Clustering.
- Hierarchical Clustering.
- DBSCAN (density-based clustering).

Phase 4: Model Evaluation & Validation

- Metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC, MSE/MAE, etc.
- Techniques: Train-Test Split, K-Fold Cross-Validation, Leave-One-Out CV (LOOCV).
- **Hyperparameter Tuning**
Grid Search, Random Search, Bayesian Optimization.

Phase 5: Deep Learning (Includes PyTorch/TensorFlow/Keras Basics)

- **Core Mechanics**
Loss Functions, Activation Functions, Gradient Descent, Forward Pass, Backward Pass (Backpropagation).
- **Neural Networks Progression**
 - Perceptron (single neuron).
 - MLP (Multi-Layer Perceptron).
 - Neural Network Architectures.
 - Classical Bottleneck in Long Sequential Data → Need for RNNs.
 - RNN, GRU, LSTM (handling sequences like time series or text).
- **Specialized Architectures**
 - CNN (Convolutional Neural Networks for images).
 - Recommendation Systems (collaborative filtering, etc.).
- **Transfer Learning**
Pre-trained models (e.g., fine-tuning on custom data).
- **Common Frameworks Setup**
Installing/running PyTorch, TensorFlow, Keras.

Phase 6: NLP & LLMs

- **NLP Fundamentals**
Tokenization, Lemmatization, Embeddings, N-gram Models, Bag of Words, Word2Vec.
- **Advanced Models** Transformers, Attention Mechanism, Self-Attention, Multi-Headed Attention.
 - Encoder vs. Decoder Architectures.
 - Autoencoders.
 - BERT, RoBERTa (bidirectional encoders).
 - GPT (generative pre-trained transformers).

Phase 7: Practical Application & Next Steps

- **Hands-on Projects**

Build a classifier, image recognizer, or sentiment analyzer.

- **Deployment Basics**

Flask/Streamlit apps, Hugging Face Spaces, or cloud inference.

- **Tools**

Jupyter Notebooks, Git for version control, Weights & Biases for experiment tracking.

- **Advanced Topics Teaser**

Generative AI (GANs, Diffusion Models), MLOps.